9. Given that $a = -i + 2j - 2k, b = 3i + 4j - 12k$. Find the angle between a and b.
A. 42° B. 32° C. 12° D. 52°
10. A body with initial velocity 135 m/s was projected at an angle 45°. What is the time taken to reach th
A. 7.8s B. 9.74s C. 9.81s D. 6.78s
11. A particle undergoes change in position from P1 to P2. This is referred to as _____
A. Kinetic energy B. Angular displacement C. Displacement D. Distance
12. An electric train moving at 15 km/h accelerates to a speed of 30 km/h in 20s. Calculate the distance in m of acceleration. A. 100m B. 125m C. 135m D. 145m
13. Determine the center of mass for three objects A,B and C on a flat board with the following masses and lo coordinate plane in meter. A: 2kg (0, 10), B: 3kg (10, 1) and C: 7kg (2, 2).
A. (3.5m, 5.5m) B. (3. 67m, 3.08m) C. (5.5m, 3.55m) D. (3.08m, 3.65m)
14. The the following statements describes an elastic collision except?
A. The body always separates after the collision B. Both momentum and kinetic energy are conserved C. Kin before collision is greater than kinetic energy after collision D. The bodies involved do not undergo any defor
15. An object in a trolley of total mass 50kg has a velocity of 5m/s. The trolley moves faster than expected for 8s increases to 9m/s. Calculate the force produced.
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ADELEKE UNIVERSITY, EDE, OSUN STATE

FACULTY OF SCIENCE, DEPARTMENT OF BASIC SCIENCES

GENERAL PHYSICS I, PHY 101 MID SEMESTER TEST

30 MINUTES



1. Which of the following represent equation of continuity
A. $\frac{A_1}{V_1} = \frac{V_2}{A_2}$ B. $A_1V_1 = A_2V_2$ C. $A_2V_1 = A_1V_2$ D. $\frac{A_1}{V_1} = \frac{A_2}{V_2}$
2. Contact angle for a fluid rising in a tube is
A. 0 degree B. Acute C. Obtuse D. 90 degrees
3. The following are the forces associated with a fluid except
A. Frictional force B. Cohesive fore C. Adhesive force D. Buoyant force
4. Dimensions helps to ..... A. reform equation B. derive equation C. all of the above D. none of the above
5. The equation $P_1 + \rho_1 g h_1 + \frac{1}{2}\rho_1 v_1^2 = P_2 + \rho_2 g h_2 + \frac{1}{2}\rho_2 v_2^2$ is known as ---------- equation.
A. Poiseuille's B. Bernoulli's C. Archimede's D. Continuity
6. The volume of the liquid passing per second (V) in a long narrow tube depends on the coefficient of viscosity (η), radius of the pipe (r) and pressure gradient (P/L), and it is illustrated by the expression $V = k\eta^a r^b (P/l)^c$, determine a, b and c
A. $a = 1,\ b = 2\ \&\ c = 3$ B. $a = 2,\ b = 1\ \&\ c = 1$ C. $a = -1,\ b = 4\ \&\ c = 1$ D. $a = -1,\ b = 4\ \&\ c = -1$
7. The viscous retarding force (F) of a spherical object moving in a viscous liquid is given by $F = k\eta^x v_t^y r^c$ where η is the coefficient of viscosity, v_t is the terminal velocity, r is the radius and k is constant. Determine x, y and z.
A. $x = 1,\ y = 1\ \&\ z = 1$ B. $x = -1,\ y = -1\ \&\ z = 1$ C. $x = 1,\ y = -1\ \&\ z = 1$ D. $x = 1,\ y = 1\ \&\ z = -1$
If the force $F_1 = 3i + 5j + 6k$ and $F_2 = 2i + 3j - 4k$ Newton are displaced by $d = 3i - 2j + 4k$ metre. Find the work done. A.
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